

CE 558: Air pollution and its mitigation

Course Code	CE 558
Course Name	Air pollution and its mitigation
L-T-P-C	3-0-0-3
Prerequisites	IC 230 (Environmental Science) for B..Tech.
Intended for	3rd and 4th year B.Tech, Post-graduate
Approval	35th BoA (Aug 2020); 28th Senate (Nov 2020), in the B.Tech Civil Engineering curriculum

Course Contents

- **The atmospheric system:** Vertical profiles of pressure and temperature; atmospheric structure; hydrostatic equilibrium and scale height; stability and lapse rates; temperature inversions; atmospheric dispersion; general circulation and surface winds; timescales of atmospheric transport. [5 Lectures]
- **Basics of atmospheric chemistry:** Photochemical and radical-assisted reactions; daytime vs nighttime and tropical vs polar chemistry; species lifetimes; atmospheric measurement units. [2 Lectures]
- **Stratospheric processes:** Dynamics of stratospheric transport; formation and distribution (latitudinal, seasonal and vertical) of stratospheric O₃; UV shielding and effect on climate; role of HO_x, NO_x and chlorofluorocarbons in catalyzing O₃ depletion- observational evidence, mitigation measures, recovery and challenges. [5 Lectures]
- **Tropospheric gas-phase species:** Local, regional and global pollutants; the oxidizing capacity of the troposphere; constraints on CO and CH₄; the CO-HO_x.NO_x.VOC-O₃ cycle; photochemical smog- case study of Los Angeles, USA; dry and wet removal of pollutants; acidic deposition - case study of the Northeastern USA; current scenario in Asia and the Indian perspective; mitigation, challenges and opportunities. [8 Lectures]
- **Aerosols:** Historical perspectives on air pollution; physical properties of aerosols - formation, growth, aging and size distribution; primary vs secondary and externally- vs internally-mixed aerosol; core-shell theory; major aerosol chemical components and their sources; tracers. [6 Lectures]
- **Air pollution control technologies:** Basics of air pollution control system design - velocities, flow rate, pressure drop, nature of combustion, acid dew point, particle settling and drag forces, diffusion; mitigation technologies in stationary systems - fluidized bed combustion, integrated gasification combined cycle, flue gas desulfurization, baghouses, scrubbers, cyclone collectors, and electrostatic precipitators; mitigation technologies in mobile systems - crankcase, evaporative and tailpipe emissions from automobiles, catalytic converters, diesel particulate filters, fuel modification and blending. [11 Lectures]
- **Risk assessment and policy intervention on air pollution:** Link between air pollution and mortality/morbidity; risk assessment for carcinogenic and non-carcinogenic pollutants; air quality indices- case study of India; air pollution as an externality; ambient, emission and technology standards; policies for air pollution control. [5 Lectures]

Text Books

1. John H. Seinfeld and Spyros N. Pandis, **Atmospheric Chemistry and Physics: From Air Pollution to Climate Change**, 3rd Edition, Wiley, 2016.
2. Daniel J. Jacob, **Introduction to Atmospheric Chemistry**, Princeton University Press.
3. Noel De Nevers, **Air Pollution Control Engineering**, 2nd Edition, McGraw-Hill, 2000.